

$$1.3.1) \bar{V}_{3 \sim 3.1} = \frac{x(3.1) - x(3)}{0.1s} = 61 \text{ m/s}$$

$$\bar{V}_{3 \sim 3.2} = \frac{x(3.2) - x(3)}{0.1s} = 60.1 \text{ m/s}$$

$$\bar{V}_{3 \sim 3.01} = \frac{x(3.01) - x(3)}{0.01s} = 620 \text{ m/s}$$

$$(2) \quad \cancel{V = \dot{x} = 20t}$$

$$V = \dot{x} = 20t \quad \therefore V_3 = 60 \text{ m/s}$$

$$(3) \quad V = \dot{x} = 20t \quad a = \dot{V} = 20$$

$$1.5. \text{ 改加速变化时刻为 } t_1, \text{ 则 } \alpha t_1 = \beta(t - t_1)$$

$$t_1 = \frac{\beta}{\alpha + \beta} t$$

$$\begin{aligned} s &= \frac{1}{2} \alpha t_1^2 + \frac{1}{2} \beta (t - t_1)^2 \\ &= \frac{1}{2} \alpha \frac{\beta^2}{(\alpha + \beta)^2} t^2 + \frac{1}{2} \beta \frac{\alpha^2}{(\alpha + \beta)^2} t^2 \end{aligned}$$

$$= \frac{1}{2} \frac{\alpha \beta}{(\alpha + \beta)^2} (\alpha + \beta) t^2 = \frac{1}{2} \frac{\alpha \beta}{\alpha + \beta} t^2$$

1.17. 假设以向下为正方向, 则下抛物球的运动为

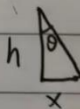
$$v_{\text{下}} = gt + v_0, \text{ 上抛为 } v_{\text{上}} = gt - v_0$$

$$\Delta v = v_{\text{下}} - v_{\text{上}} = 2v_0$$

$$\therefore S = \Delta v t = 2v_0 t$$

$$1.21. \quad h = \frac{1}{2} g t^2, \quad x = vt,$$

$$\therefore \theta = \arctan \frac{x}{h} \approx 16^\circ 45'$$



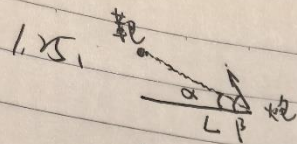
$$1.24. \text{ 设抛射角为 } \theta, \text{ 则 } v_x = v_0 \cos \theta, \quad v_y = v_0 \sin \theta$$

$$\text{碰前 } t_1 = \frac{5}{v_x}, \quad v_{y1} = v_y - g t_1, \quad y_1 = v_y t_1 - \frac{1}{2} g t_1^2$$

$$\text{碰后 } t_2 = \frac{v_{y1}}{g}, \quad | \Delta x | = v_x t_2 = 5 \text{ m}$$

$$\text{联立上述式子解得 } \sin 2\theta = \frac{20g}{v_0^2} \text{ 代入 } g, v_0$$

$$\text{解得 } \sin 2\theta = \frac{49}{64} \approx 25^\circ \text{ 或 } 65^\circ$$



$$h = L \tan \alpha, \quad t = \frac{L}{v_0 \cos \beta}, \quad v_0 \sin \beta t - \frac{1}{2} g t^2 = h$$

$$\text{解得 } v_0 = \frac{\sqrt{L g \cos \alpha}}{2 \cos \beta (\sin \beta - \alpha)}$$

1.31 (1) 此时 $a_n = a_t = 3 \text{ m/s}^2$

而

$$a_t = \frac{dv}{dt}, \quad a_n = \frac{v^2}{R}, \quad \text{联立得 } v = a_t t$$

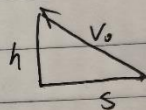
$$\text{解得 } t = 1 \text{ s}$$

(2) 将圆周看作直线的匀加速

$$s = \frac{1}{2} a_t t^2 = 1.5 \text{ m}$$

1.32 (1) $l = \sqrt{h^2 + s^2}$

$$\frac{dl}{dt} = -v_0$$



$$l^2 = h^2 + s^2 \quad \text{两边对 } t \text{ 求导}$$

$$2l \frac{dl}{dt} = 2s \frac{ds}{dt}$$

$$\therefore -l v_0 = s \frac{ds}{dt} \quad \therefore v = \frac{ds}{dt} = -\frac{l}{s} v_0 = -\frac{\sqrt{s^2 + h^2}}{s} v_0$$

$$\text{即 } v \text{ 的大小为 } \frac{\sqrt{s^2 + h^2}}{s} v_0$$

$$\text{对 } v = \frac{ds}{dt} = -\frac{l}{s} v_0 \text{ 求导}$$

$$\text{则 } a_s = \frac{dv}{dt} = -v_0 \cdot \frac{-v_0 s - v l}{s^2} = -v_0 \frac{v_0 s + \frac{l^2}{s} v_0}{s^2}$$

$$= -\frac{v_0^2 s^2 - l^2 v_0^2}{s^3} = -\frac{v_0^2 h^2}{s^3}$$

$$\therefore a \text{ 的大小为 } \frac{v_0^2 h^2}{s^3}$$

$$2.1. \quad m_B g - 2m_A g = (m_A + m_B) a$$

$$\therefore a = \frac{m_B g - (m_A + m_B) g}{m_B g}$$

2.7, 记绳的拉力为 T , 则

$$T \cos \frac{\alpha}{2} = m_2 g \cos (\pi - \theta - \frac{\alpha}{2})$$

$$T \cos \frac{\alpha}{2} = m_1 g \cos (\theta - \frac{\alpha}{2})$$

$$\text{即 } m_2 \cos (\theta + \frac{\alpha}{2}) + m_1 \cos (\theta - \frac{\alpha}{2}) = 0$$

$$\text{整理得 } \cos \theta \cos \frac{\alpha}{2} (m_1 + m_2) = \sin \theta \sin \frac{\alpha}{2} (m_2 - m_1)$$

$$\text{即 } \tan \theta = \cot \frac{\alpha}{2} \frac{m_1 + m_2}{m_2 - m_1}$$

2.8, 当脱离时, 弹力恰好提供 A, B 加速度为 0, 即恰好为平衡位置

$$\text{弹力 } F = kx = m_1 \ddot{x} = m_2 \ddot{x}$$

$$\text{质量 } m = m_A + m_B = 0.2 \text{ kg} \quad \frac{1}{2} k x_0^2 = \frac{1}{2} m v^2 \quad \therefore v = x_0 \sqrt{\frac{k}{m}}$$

$$\text{另解 } x(t) = x_0 \cos(\omega t) \quad \omega = \sqrt{\frac{k}{m}} = 200 \text{ cm/s}$$

$$(2) \quad E_A = \frac{1}{2} m_A v^2 = \frac{1}{2} k x^2$$

$$\text{解得 } x = 5.2 \text{ cm}$$

$$2.10, \text{ 水平方向上 } ma = N \sin \theta$$

$$\text{竖直方向上 } mg = N \cos \theta$$

$$\therefore a = g \tan \theta$$

$$\therefore F = (M + m) g \tan \theta$$

2.16 (1) 对M分析, 沿斜面方向 $Mg \sin \alpha + N \sin \alpha = Ma_m$

其中 N 是 m 与 M 之间的压力

对 m 分析, $mg - N = ma_m \Rightarrow N = m(g - a_m)$

由于无摩擦, m 只有向下的加速度

$$a_m = a_M \sin \alpha$$

$$\therefore N = m(g - a_M \sin \alpha)$$

$$\text{解得 } a_M = \frac{(M+m)g \sin^2 \alpha}{M + m \sin^2 \alpha}$$

(2) ~~对M在竖直方向上分析, $Mg + N \cos \alpha = Ma_M$~~

~~又对M在水平方向上分析, $N \sin \alpha = Ma_M$~~

对 ~~M和m~~ 整体在竖直方向上

$$(M+m)g - T \cos \alpha = (M+m)a_m$$

$$\therefore T = \frac{M(M+m) \cos \alpha}{M + m \sin^2 \alpha}$$

2.17 设绳拉力为 T 则 $T - m_2g = m_2a$

斜面支持力为 N 则 $m_1g \cos \alpha = N$

$$m_1g \sin \alpha - T = m_1a$$

联立求解 ~~$T = \frac{m_1 m_2 g \sin \alpha}{m_1 + m_2}$~~

$$a = \frac{m_1 \sin \alpha - m_2}{m_1 + m_2} g$$

~~又对斜面上的 $F =$~~

F 提供整体质心的向右加速度

$$\text{即 } F = m_1 a \cos \alpha$$

$$= \frac{m_1 \sin \alpha - m_2}{m_1 + m_2} m_1 g \cos \alpha$$

$$22) \quad \vec{f} = -k m \vec{v}$$

$$x \text{ 轴向上 } m \frac{dv_x}{dt} = -k m v_x \Rightarrow \int_x dv_x = -k dt$$

$$y \text{ 轴向上 } m \frac{dv_y}{dt} = -mg - k m v_y \quad (\text{取向上为 } \vec{z})$$

$$v_x(0) = v_y(0) = \frac{\sqrt{2}}{2} v_0$$

$$\ln v_x = -kt + C$$

$$\therefore v_x = e^{-kt} \cdot e^C \cdot \text{const} \cdot v_x(0)$$

$$\text{得 } v_x = \frac{\sqrt{2}}{2} v_0 e^{-kt}$$

$$\text{同理 } v_y = \left(\frac{\sqrt{2}}{2} v_0 + \frac{g}{k} \right) e^{-kt} - \frac{g}{k}$$

$$\text{令 } v_x = -v_y$$

$$\text{解得 } t = \frac{1}{k} \ln \left(\frac{\sqrt{2} k v_0 + g}{g} \right)$$

$$S = \int_0^t v_x(t) dt$$

$$= \frac{\sqrt{2}}{2} v_0 \cdot \frac{1 - e^{-kt}}{k}$$

$$\text{而 } e^{-kt} = \frac{g}{\sqrt{2} k v_0 + g}$$

$$\therefore S = \frac{v_0^2}{\sqrt{2} k v_0 + g}$$